INCH-POUND

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PERFORMANCE SPECIFICATION

LARGE CAPACITY HIGH PRESSURE BREATHING AIR COMPRESSOR (HPBAC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the High Pressure Breathing Air Compressor (HPBAC) used by Navy personnel as part of the shipboard breathing air system to refill breathing air cylinders used with the open-circuit self contained breathing apparatus (SCBA). This equipment will be used on Navy ships and at Navy Training Centers for rapid charge of breathing air cylinders.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment, and Systems, Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Influence Characteristics of Subsystems and Equipment

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7.1 - American National Standard, Compressed Gas Association. Commodity Specification for Air.

(Copies of this document are available from the Compressed Gas Association, Inc., 4221 Walney Road, 5th Floor, Chantilly, VA 20151-2923 or online at www.cganet.com)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specification exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The contractor shall select materials capable of meeting all of the operational and environmental requirements specified herein.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Corrosion resistance. The HPBAC shall be constructed from corrosion resistant materials due to exposure of shipboard environment.

3.3 Design.

3.3.1 Size. The overall size of the HPBAC shall not exceed 50 inches wide by 50 inches deep by 65 inches high.

3.3.2 Weight. The HPBAC shall have a dry weight, not including resilient mounts, of less than 1700 lbs.

3.3.3 Workmanship. The HPBAC and all accessories shall be free from damage such as cracks, burrs, sharp edges, tears, punctures, mars, abrasions, pits and any other defects that will adversely affect the performance, reliability, safety, durability, or appearance of the HPBAC. The HPBAC and all accessories shall be free from foreign material such as dirt, grease, oil, or metal chips. Rubber parts shall be compact (free from holes or bubbles) and free from foreign matter. Metal parts shall be free of distortion, burrs, sharp edges, scratches, or other damage. No voids shall be evidenced in any metal-to-rubber bond areas.

3.3.4 Safety. To ensure maximum protection for operating and maintenance personnel against hazards and accidents, the compressor shall incorporate the following safeguards and protective features:

a. Exposed moving parts shall be covered by removable guards of expanded metal or similar sturdy serviceable metal.

b. Exposed component surfaces with an operating surface temperature in excess of 125 ºF shall be shielded to prevent human contact during HPBAC operation. Shielding shall be removable to permit HPBAC maintenance and repair.

c. Piping under pressure and attached to the specified compressor shall be safeguarded against sudden and catastrophic rupture by hydrostatic testing and nondestructive testing of welds and joints.

d. Exposed edges shall be rounded and made smooth to prevent cutting edges and sharp corners.

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3.3.5 Interchangeability. Part and component interchangeability shall be inherent in the HPBAC design. All parts and assemblies assigned a unique part number shall be functionally and physically interchangeable between the same model HPBACs.

3.3.6 Maintenance access. The HPBAC shall be configured to permit access to internal components for inspection and maintenance. The use of removable panels is acceptable.

3.3.7 Resilient mounts. Shock and/or vibration isolating resilient mounts shall be provided for mounting the HPBAC on the foundation. Mounting bolt size, grade, and arrangement shall be indicated.

3.4 Performance characteristics.

3.4.1 Air delivery. The HPBAC shall deliver air for charging breathing air cylinders to a nominal pressure of 4500 psig. The HPBAC shall be capable of delivering breathing air at a maximum of 5000 psig. HPBAC shall include adjustable over pressurization devices to prevent over pressurizing breathing air cylinders during charging. The compressor shall deliver a minimum of 20 cubic feet per minute (CFM) of free air at a delivery pressure of 3000 psig.

3.4.1.1 Cooling. The compressor shall be air-cooled. The cylinders shall be arranged such that adequate air cooling is provided. The cylinders shall be removable from the crankcase. A cooling fan shall be provided and the fan wings shall be replaceable.

3.4.1.2 Lubrication. The compressor shall be oil lubricated or an oil-free reciprocating compressor. Compressor designs, which use oil lubrication, shall employ either a splash or pressure lubrication system. A pressure lubrication system, if used, shall include a positive displacement oil pump and a pressure regulator. A means to check the oil level shall be provided, such as a dipstick or sight glass. The compressor oil drain shall be piped to prevent oil spillage on the HPBAC when the compressor is drained.

3.4.1.3 Coalescing separator. A separator shall be supplied after each stage of compression, if necessary, to provide adequate moisture removal. A coalescing separator shall be supplied after the final stage of compression.

3.4.1.4 Automatic condensate drain (ACD) system. An automatic drain system shall be supplied for all separators. Manually operated valves shall be supplied to override the automatic operation of the drain system for test and maintenance purposes. An exhaust muffler and condensate reservoir shall be supplied on ACD system.

3.4.1.5 Inlet air filter. The compressor shall be equipped with an inlet air filter with replaceable particulate element. An inlet air filter maintenance indicator shall be provided on the instrument panel.

3.4.2 Electric drive motor. The HPBAC shall be driven by a 440-VAC, 60-Hz, 3-phase, electric motor. The electric motor shall be of the open drip-proof (ODP) design. If power from the motor will be transmitted to the compressor by a v-belt drive, the v-belt drive shall be designed to tighten the drive belts automatically. An arrow to indicate the rotational direction of the compressor rotor and the electric motor shaft shall be indicated in locations that are viewable without requiring disassembly of the compressor.

3.4.3 Control panel.

3.4.3.1 Selector switch. An ON/OFF selector switch shall be provided on the instrument panel to control power to the HPBAC. Selector switches shall be of the maintained contact type. Turning the ON/OFF selector switch to the “OFF” position shall remove all electric power from the compressor motor, the compressor terminal boards, the air pressure switches, and all control and surveillance instruments and devices.

3.4.3.2 Controls. The HPBAC shall have controls which permit automatic starting and stopping of the compressor based on the outlet air pressure. The control setpoints shall be user adjustable and initially set at approximately 4500 psig (start compressor) and 5000 psig (stop compressor).

3.4.3.3 Pushbuttons. “ON” and “OFF” pushbuttons shall be provided for starting and stopping the compressor. Depressing the “ON” pushbutton shall start the compressor, and the system shall automatically start and stop to maintain the system’s operating pressure between the high and low setpoints of the system. Depressing the “OFF” pushbutton shall stop the compressor.

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3.4.3.4 Panel. The panel shall indicate the operational status of the HPBAC, the air pressure at each stage, the compressor oil pressure, an indication of the inlet air filter condition, and all fault alarm conditions.

3.4.3.5 Hourmeter. A non-resettable hourmeter shall be supplied to record the number of compressor run hours. The hourmeter shall have an elapsed run time meter, incremented in hours, and shall be installed on the instrument panel.

3.4.3.6 Oil pressure. The HPBAC's oil pressure shall be monitored by a pressure switch and pressure gauge installed on the instrument panel. The system shall shut down and a fault light illuminate should the compressor's oil pressure drop below the preset value during operation. The oil pressure switch shall be by-passed during start-up to permit the oil pump to achieve the normal operating pressure.

3.4.3.7 Temperature switch. A temperature switch shall be supplied on the discharge line of the final stage of compression. The system shall shut down and a fault light illuminate should the final discharge temperature exceed the tamper-proof set point during operation.

3.4.3.8 Fault lights. All fault lights shall be installed on the instrument panel and shall be red in color. Each fault light shall be identified by an engraved nameplate.

3.4.4 Purification system. The HPBAC shall have a purification system that includes an oil-water separator, a condensate drain valve, and a breathing air purifier.

3.4.4.1 Air quality. The HPBAC purification system shall deliver Grade D breathing air in accordance with ANSI/CGA G-7.1. The Grade D air shall have a maximum moisture content of 20 mg/m^3 or a dew point less than or equal to -65 ºF.

3.4.4.2 Processing capability. The breathing air purifier shall have a processing capability of at least 200,000 standard ft^3 of air based on an ambient inlet air temperature of 70 °F and a relative humidity of 95%.

3.4.4.3 Working pressure. The purification system shall have a working pressure of 5000 psig.

3.4.4.4 Cartridges. Purification system cartridges shall be replaceable without disconnecting purification system piping. Purification chambers shall preclude operation of the HPBAC when purification system cartridges are not installed or are improperly installed. The HPBAC shall shut down automatically should the operator fail to change any cartridges within the warning period. The system shall not be capable of restarting until any saturated cartridges are replaced with new ones.

3.4.4.5 Bleed valve. A bleed valve shall be provided to vent the purification system to facilitate maintenance.

3.4.4.6 Pressure regulating and check valves. A pressure regulating valve and a check valve shall be supplied downstream of the purification system to increase the efficiency of the purification system by maintaining a positive backpressure. A check valve shall be supplied between the coalescing separator on the compressor's discharge line and the purification system to maintain the positive pressure in the purification system when the compressor shuts down.

3.4.4.7 Display module. A display module shall be mounted to the HPBAC to indicate the status of each purification system cartridge. When electrical contact between the display module and any of its sensors is lost, the HPBAC shall cease to operate, and a fault shutdown shall be indicated on the display module by an illuminated fault light.

3.4.4.8 Moisture monitor. A sensor shall be located in the purification system for direct monitoring of the moisture level. The moisture monitoring system shall warn the operator, in advance, of the impending saturation of any purification cartridges. The moisture monitoring system shall be of a fail-safe design. A manual override shall be supplied for the moisture monitor.

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3.5 Environmental conditions.

3.5.1 Operating temperature. The HPBAC shall be fully operational at an ambient temperature between 32 °F and 115 °F.

3.5.2 Air outlet temperature. The HPBAC delivered outlet air temperature shall be not more than 20 °F above the ambient temperature.

3.5.3 Shock. The HPBAC shall meet the Grade A shock acceptance requirements of MIL-S-901 when tested as specified in 4.5.4.

3.5.4 Vibration. The HPBAC shall meet the Type I vibration requirements of MIL-STD-167-1 when tested as specified in 4.5.5.

3.5.5 Electromagnetic interference (EMI). The HPBAC shall meet the acceptance criteria specified in MIL-STD-461 when tested as specified in 4.5.6.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete HPBAC when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.6 (see Table I).

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.3 (see Table I).

4.4 Examination. Each HPBAC shall be examined for compliance with the requirements specified in 3.2 through 3.5. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE I. First article and conformance inspections.

Test	Requirement Para.	1st Article	Conformance
Hydrostatic	3.3.4	4.5.1	4.5.1
Capacity	3.4.1	4.5.2	4.5.2
Performance	3.4	4.5.3	4.5.3
Shock	3.5.3	4.5.4	
Vibration	3.5.4	4.5.5	
EMI	3.5.5	4.5.6	

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4.5 Methods of inspection.

4.5.1 Hydrostatic. All piping components and assemblies which will be subjected to normal working pressures above 30 lb/in^2 shall be hydrostatically pressure tested to 1½ times the associated maximum design working pressure. The test shall consist of applying demineralized water under steady pressure for a period of not less than 30 minutes. Tap water may be used in lieu of demineralized water if the chloride content does not exceed 5 ppm. During the interval that the component is under pressure, the manufacturer shall inspect each piping component and assembly for strength and porosity weakness as indicated by leakage, excessive sweating, and material deformation or rupture. If the component or assembly does not exhibit leakage, excessive seating, deformation or rupture, it shall be considered acceptable.

4.5.2 Capacity. The HPBAC shall be subjected to a capacity test to ensure compliance with 3.4.1 of this specification.

4.5.3 Performance. Each production unit shall be subjected to a performance test to ensure that the HPBAC meets all performance requirements, and that all systems, controls, safety, surveillance and monitoring devices operate as specified herein. The performance test shall be an operational test of forty hours duration. The forty hours of HPBAC operation shall not be continuous, but shall consist of five 8-hour runs with rest periods of not less than four hours between each run. Artificial casualty conditions shall be imposed at the beginning and end of the forty-hour performance test to ensure satisfactory operation of the automatic shutdown features. If an oil filter is used, a disposable type shall be replaced, and a permanent-type shall be cleaned, following the forty-hour performance test. When specified (see 6.2), test data shall be provided.

4.5.4 Shock test. One HPBAC shall be subjected to a Grade A, heavy weight, class II, type A shock test in accordance with MIL-S-901. HPBAC shall be qualified deck mounted and for unrestricted orientation. Shock testing shall be accomplished with the as-supplied resilient mounts. The HPBAC shall meet the acceptance criteria specified in 3.5.3 after the test.

4.5.5 Vibration test. One HPBAC shall be subjected to a Type I vibration test in accordance MIL-STD-167-1. The HPBAC shall meet the requirements of 3.5.4 after the test.

4.5.6 EMI test. One HPBAC shall be subjected to the EMI tests specified in MIL-STD-461 requirement matrix for surface ship equipment and subsystems, in accordance with the test procedures of MIL-STD-461. Acceptance criteria shall be in accordance with 3.5.5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The compressors covered by this specification are intended for compressing air to recharge breathing air cylinders used in shipboard fire fighting applications. They are not for potable water use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required. (see 3.1)

c. Test data requirements. (see 4.5.3)

d. Packaging requirements. (see 5.1)

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6.3 Subject term (key word) listing.

Air charging system

Breathing air supply

Emergency air

SCBA

Custodians:
Army – EA
Navy – SH
Air Force – 03

Preparing Activity:
Navy – SH
(Project 4240-0698-000)

Review Activities:
Navy – AS
Air Force – 11, 84, 99

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